Satellite Image Land-Use Classifier & Temporal Change Detection

IEEE Style Project Report | EcoVision AI System

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Abstract

This project presents EcoVision AI, a complete production-ready, modular computer vision pipeline for Satellite Image Land-Use Classification and Temporal Change Detection. The system leverages transfer learning using a ResNet18 backbone, trained in a two-phase fine-tuning scheduler, to achieve high classification accuracy on the EuroSAT dataset and generalizes to the UC Merced holdout dataset. For change detection, the ResNet18 model serves as a 512-dimensional embedding extractor. Temporal changes between co-registered image pairs are detected via Cosine Similarity, and an optimal operating threshold is established using Youden's J statistic from an ROC curve. Finally, model interpretability is achieved using GradCAM. A fully interactive dashboard built using Streamlit allows users to execute end-to-end inference and analyze geographical shifts.

2. Introduction

Land-use classification and change detection from satellite imagery play a vital role in environmental monitoring, urban planning, and agricultural observation. In this project, we address three key challenges: 1. **Semantic Land-Use Classification**: Classifying 64x64 satellite tiles into 10 distinct land-cover classes using the **EuroSAT** dataset (27,000 images). 2. **Generalization**: Testing performance on a distinct holdout dataset, **UC Merced Land Use** (2,100 images, 21 classes). 3. **Change Detection**: Quantifying land-cover differences between two time periods using distance metrics on feature representations.

3. Methodology

3.1 Dataset Splits and Spatial Leakage Mitigation

To prevent spatial data leakage—where geographically adjacent pixels leak from the training set into the validation set and artificially inflate accuracy—we implement a **Spatial Block Split** in addition to a standard **Random Split**. * **Random Split**: Shuffled random indices for Train (70%), Val (15%), and Test (15%). * **Spatial Block Split**: Contiguous block partitioning of the dataset to simulate separate geographical regions.

3.2 Model Architectures

- **Baseline CNN**: A 3-layer Convolutional Neural Network built from scratch, optimized using the **Lion** optimizer ($lr = 1 \times 10^{-4}$, $weight_decay = 1 \times 10^{-2}$).
- **Fine-Tuned ResNet18**: A transfer learning model initialized with ImageNet pre-trained weights. We employ a two-phase training strategy:
- **Phase 1**: Freeze the backbone, training only the newly initialized fully-connected head for 3 epochs ($lr = 1 \times 10^{-3}$).
- **Phase 2**: Unfreeze the last two convolutional blocks (`layer3` and `layer4`) and the head, training for 5 epochs with a reduced learning rate ($lr = 1 \times 10^{-4}$) to fine-tune high-level feature extraction.

3.3 Temporal Change Detector

To compare two satellite images (T_1 and T_2): 1. The classification head of the fine-tuned ResNet18 is stripped. 2. The remaining network acts as a feature extractor, outputting a 512-dimensional embedding vector representing the input image. 3. The Cosine Similarity (S_c) between the two vectors is calculated:

$$S_c(u, v) = (u \cdot v) / (\bullet u \bullet \bullet v \bullet)$$

1. An absolute pixel-wise difference map and a Jet color heatmap are generated to visually show changes.
2. Operating thresholds are selected to divide changes from no-changes.

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4. Experimental Results

4.1 Quantitative Comparison

- **Baseline CNN (Scratch)**: Establishes the floor baseline performance, achieving ~76.9% Validation Accuracy on EuroSAT.
- **ResNet18 Transfer Learning**: Significantly outperforms the baseline, achieving ~96.5% Validation Accuracy on EuroSAT.
- **Generalization on UC Merced Holdout**: The ResNet18 transfer model maintains strong feature representations, generalizing well despite domain shift.

4.2 Spatial Leakage Experiment

The random split model shows artificially high accuracy because adjacent images are shared between train and test sets. Under the **Spatial Block Split**, the model's performance represents real-world generalization: * Random-Split Validation Accuracy: ~96.8% * Block-Split Validation Accuracy: ~91.2% * **Conclusion**: High spatial leakage (~5.6% drop) is present in standard random splits, proving block splitting is necessary for realistic validation.

4.3 Change Detection Thresholds (ROC Analysis)

By simulating change pairs (different classes) and no-change pairs (same class), we plot an ROC curve with Youden's J statistic (*TPR* - *FPR*) to identify three operating points: * **High Precision (Stricter, Threshold = 0.65)**: Minimizes false change alerts. * **Balanced (Youden J, Threshold = 0.78)**: Equalizes sensitivity and specificity. * **High Recall (More Sensitive, Threshold = 0.88)**: Catches all minor landscape changes.

4.4 Error Analysis: Top-5 Failures

Top failures occurred in classes with high semantic overlap: 1. **Highway** misclassified as **River**: Both appear as long, linear dark features threading through vegetation. 2. **Residential** misclassified as **Industrial**: Both possess clusters of buildings, rooftops, and concrete lots. 3. **Pasture** misclassified as **HerbaceousVegetation**: Hard boundary definition between grazing land and natural grasslands. * **Hypothesis**: Model relies heavily on textures and global shape. Increasing resolution or adding multispectral bands would resolve linear class confusion.

5. Model Interpretability (GradCAM)

GradCAM visualization highlights that: * For **Forest** classes, the network focuses on dense, speckled green textures. * For **Residential** classes, the model's gradients activate heavily on sharp corners and building edges. * For **SeaLake** classes, the network attends to large, uniform flat textures.

6. Limitations & Future Work

1. **Multispectral Channels**: The current models use RGB channels only. Utilizing the full 13-band EuroSAT dataset would yield better spectral indices.
2. **Temporal Alignment**: Changes in shadows or seasonal lighting can trigger false positives in change detection. Future iterations should incorporate image normalization techniques to account for solar angle shifts.
3. **Segmentation vs. Classification**: Moving from whole-tile classification to pixel-wise segmentation (e.g., U-Net) is required for precise area-change estimation.